

MATH 485 Final Project Proposal

Weather Forecasting Ensembles for Prediction Market Trading

Introduction

This project develops an ensemble based machine learning technique that will generate weather forecasts in order to find market inefficiencies on Kalshi's weather prediction market. Kalshi hosts a variety of weather prediction markets, usually focused on daily temperature highs/lows in certain cities, along with other markets featuring rain, snow, monthly records, natural disasters, and climate goals. We plan on building a system that will use data from multiple public weather sources that will train a stacked ensemble of models (probably Random Forest, XGBoost) to produce calibrated probability estimates.

The pipeline will begin with feature engineering and dimensionality reduction using RFE and PCA. We will then train and compare individual models before combining them into a stacked ensemble. When the ensemble's predicted probability diverges sufficiently from the market probability, we will flag it as a trading opportunity. We will use a straightforward Kelly Criterion based threshold strategy to determine position sizing, but may automate it using a MDP in the future.

Expected Outcome

We expect a fully functional pipeline that ingests live weather data and outputs calibrated probability predictions for temperature and precipitation events. Along with this, we will use a stacked ensemble featuring Random Forest and XGBoost models that will allow us to compare our predictions with the market predictions, giving us access to expected-value opportunities. We will validate the system through backtesting over historical Kalshi contracts, along with simulating trading performance and computing profit/loss metrics.

Initial Plan

- Research and Data Preparation

- By April 14, we plan on completing the initial weather data and Kalshi research. This research will include aggregating multiple weather data sources, determining which APIs best cover the cities and event types on Kalshi, and conducting exploratory data analysis.
- Design and Prototyping
 - By April 30, we plan on implementing the feature engineering portion. We also plan on training at least a Random Forest and XGBoost model (maybe future class topics) and tuning hyperparameters via cross-validation. We will be designing and training the stacking ensemble, and developing the trading signal module that compares our ensemble probability with the Kalshi market data.
- Experimentation
 - By May 7, we plan on simulating the model across the historically completed contracts. We will compute evaluation metrics along with building some visualizations and summary statistics to get a better grasp of the model.

Task Per Team Member

- We plan on all helping out where we can, but we have focused areas for each member:
 - Derrick Jones
 - Sourcing the data from multiple existing sources, collecting historical Kalshi contract data, and cleaning/merging datasets and conducting initial exploratory analysis.
 - Rowan Morkner
 - Feature engineering, training and tuning of individual models, and designing the stacked ensemble.
 - Harrison Lloyd
 - Developing the signal for when a market inefficiency is calculated and implementing evaluation metrics for fine-tuning the performance of our model